Best Practices for Lego Prototypes

Lego is a staple of any kid's toy box. Its versatility and ability to spark the imagination is what drives the company's success. As a designer, you can take advantage of Lego's versatility to create quick and simple prototypes of your ideas. The best part of using Lego pieces to build your prototypes is that they become easy to dismantle and tweak; simply detach a part of your Lego prototype, swap it with an alternative design, and play with it to see if it works.

Tim Brown, CEO of international design firm IDEO, recounts in his book *Change by Design* that Lego prototyping has been widely used in IDEO's design thinking process. The design team at IDEO even used it to create a prototype for a complex insulin injection device!

Pros and Cons of Lego Prototypes

Pros of Lego prototypes

- Lego bricks allow you to **quickly create physical prototypes**—you can build a rough model faster than most 3D printers can! They allow you to produce relatively cheap prototypes for products that are physical and tangible.
- Lego prototypes are **versatile and easy to modify and dismantle**. You can easily remove, add or rotate bricks to change your prototype.
- Lego prototypes **encourage experimentation and fun**, which are important components of success in the design thinking process.

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Cons of Lego prototypes

- Lego prototypes are **not so suitable for intangible products**, such as mobile apps or websites. However, they can still be used to create user journey stories for such intangible products.
- Lego prototypes are **relatively expensive** low-fidelity prototypes—especially if you don't have a set of Lego bricks. In that case, you of course have to first purchase some bricks, which cost more than other forms of prototypes, such as paper prototypes.

When to Use Lego Prototypes

- Use Lego prototypes to **empathize** with your users. Use Lego bricks to recreate and reenact user journeys cheaply and visually.
- Use Lego prototypes when your solution involves a **complex system of different parties**. You can use different Lego characters to represent each party involved, so that you don't miss any of their needs in the final product.
- When you're creating **complex physical products**, you can use Lego prototypes to create quick and dirty prototypes. Use these to get a rough sense of how large or heavy the final product will feel.

Best Practices and Tips for Lego Prototypes

- Like sketches, Lego prototypes don't require any level of artistic talent. Use this as an opportunity to **involve your non-designer team-mates and stakeholders**.
- Lego prototypes are best used to **reenact user journeys**. Use Lego characters to run through what a day is like for your user—this is a great empathy-building exercise for your team!
- Use Lego prototypes to **mimic the actual size** of a proposed physical product. This way, for example, you can test whether it fits into your jeans pocket.
- You can also use Lego prototypes to **mimic the actual weight** of your proposed product. Since the Lego bricks are pretty light, remember that you can always place weights into your Lego creations to add more heft to them!

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